

HAN2026: Risk-Adaptive Bilateral Negotiation Agent

Agent Name: T2Agent
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Abstract. T2Agent heuristically estimates negotiation breakdown risk from observable interaction signals and adjusts its bidding and acceptance criteria accordingly. For human-facing negotiation, offer selection and acceptance remain numerical, while a language model is used only to produce understandable utterances. This separation preserves strategic stability and interpretability while allowing natural communication.

1. Introduction

T2Agent adjusts its bidding and acceptance criteria as negotiation risk changes. It estimates risk heuristically from elapsed time and the opponent’s concession behavior, then uses that estimate to set a dynamic own-utility target. Offer selection and acceptance are determined numerically, while the language model is restricted to understandable utterance generation. This division provides both strategic stability and explainability.

2. The Design of T2Agent

At each turn, the opponent’s offer is added to the opponent model, breakdown risk is estimated, and a corresponding own-utility target is calculated. Candidate offers are evaluated with a near-Pareto filter. The agent accepts an opponent offer when it satisfies the acceptance rule; otherwise, it selects a suitable counteroffer near the target. If the target cannot be reached, the search is relaxed in stages. A short utterance is generated only after the strategic action has been fixed.

Observe offer	→	Estimate model and risk	→	Accept or select bid
Language generation after the strategic decision				

After the numerical decision, the model receives the desired tone, selected issue–value pairs, inferred opponent-priority issue, main concession, changed values, and latest opponent utterance. It receives no utilities, risk, candidate scores, or strategy parameters. The prompt requests a short cooperative but appropriately firm message without changing the offer.

We use $u(o)$ for normalized own utility, $\hat{v}(o)$ for estimated opponent utility, ρ for normalized reservation utility, b_t for breakdown risk, T_t for the proposal target, and τ for relative negotiation time.

3. Concealing Bidding Strategy

Let σ denote a safety margin above the reservation utility. The raw own-utility target is

$$T_t^{\text{raw}} = \max(\rho + \sigma, 1 - b_t).$$

This equation serves two purposes. Subtracting risk from the top of the normalized utility scale directly converts greater breakdown danger into greater concession. The reservation floor with σ prevents risk adaptation from producing a bid that is not sufficiently beneficial to T2Agent.

When an opponent concession reduces risk, the target may rise again. To avoid abruptly retracting an earlier concession, the upward change is limited by κ :

$$T_t = \min(T_t^{\text{raw}}, T_{t-1} + \kappa).$$

The purpose of this asymmetric constraint is to preserve behavioral continuity: the agent does not suddenly reclaim value it has already conceded, but it can still lower its target promptly when breakdown risk rises.

Every outcome is evaluated in the two-dimensional space $(u, \hat{v})$. A near-Pareto filter with tolerance ε removes an outcome when another is nearly no worse for both parties and clearly better for at least one. Among frontier offers above T_t , the policy first minimizes distance to the target, then prefers higher estimated opponent utility and greater novelty.

The tolerance is included because the opponent model is approximate. It removes clearly inefficient offers without treating small estimation differences as reliable evidence of dominance.

If no frontier offer reaches the target, it is lowered by a relaxation amount δ_T . Search then expands successively to the whole near-Pareto set, all outcomes above the reservation margin, and all individually rational outcomes. Broader fallback tiers use

$$S(o) = w_o \hat{v}(o) + w_{t(1-|u(o)-T_t|)} + w_n N(o),$$

where $N(o)$ is the minimum fraction of changed issue values relative to recent bids. The three terms respectively reward opponent acceptability, adherence to the intended concession level, and proposal novelty. Their combination is used to avoid improving one consideration while ignoring the others. Exact repetitions within a recent-offer window are excluded during normal selection. They are permitted only if no non-repeated individually rational bid remains, so novelty does not prevent a viable agreement when the outcome space is small.

4. Acceptance Strategy

An accepted offer must always exceed the true reservation utility. The agent first compares the received offer with the denormalized form of

$$A_t = \max(\rho, 1 - b_t).$$

This keeps acceptance demanding at low risk while gradually relaxing it as the probability of disagreement rises.

Before rejecting, T2Agent also computes its planned counteroffer and accepts whenever

$$U(o_{\text{received}}) + \eta \geq U(o_{\text{next}}).$$

This prevents the agent from rejecting an offer that is already at least as good as its intended continuation. The tolerance η absorbs small differences caused by discrete outcomes or numerical evaluation. If no counteroffer strictly above the reservation value exists, the agent ends the negotiation. In this case, every available outcome is unacceptable to T2Agent, so continuing the negotiation cannot produce an individually rational agreement.

5. Opponent Model

The opponent model learns exclusively from observed offers. For each issue value it stores an exponentially decayed frequency count with factor λ , so recent and repeated values receive larger scores. Issue importance is inferred from stability: issues changed less often by the opponent obtain greater weight. Until sufficient observations are available, issue weights remain uniform. Decay is used to adapt when the opponent changes behavior, while the uniform initial model avoids drawing a strong conclusion from scarce evidence.

Estimated opponent utility combines a history score, similarity to the first and latest offers, and a weak inverse prior $1 - u(o)$. With mixture weights that sum to one, the estimate is

$$\hat{v}(o) = w_h H(o) + w_a A(o) + w_{i(1-u(o))}.$$

The history term learns repeated preferences, the anchor term preserves evidence from the opponent's stated bargaining positions, and the inverse prior supplies a usable estimate before enough observations exist. As evidence accumulates, weight is shifted toward history so that the data gradually replaces the prior.

Recent changes in the own utility of opponent offers are also used to classify behavior as conceder, brawler, cooperative-human, volatile, or hardliner. The profile controls

$$b_{\text{time}} = \tau^p.$$

The exponent adapts the urgency curve to the observed profile. Volatile and hardliner profiles use an exponent below one, causing time risk to increase earlier. The cooperative-human profile uses an exponent of one. Conceder and brawler profiles use an exponent above one, allowing the agent to preserve its target for longer.

The final risk combines time pressure, opponent concession behavior, and offer gap. Let Δ_t be the latest offer's own-utility improvement relative to a recent reference window. The offer gap is defined as the absolute difference

$$G_t = |\hat{v}(o_t^{\text{opp}}) - \hat{v}(o_{t-1}^{\text{self}})|.$$

It measures the distance, under the opponent model, between the latest opponent offer and T2Agent's previous bid. The final risk is

$$b_t = \text{clip}(\tau^p + \text{clip}(-k_c(\Delta_t - \delta_c), -l_d, l_u) + k_g G_t).$$

The time term represents unavoidable deadline pressure. The concession term reduces risk after meaningful movement and raises it after stagnation or deterioration; its bounds prevent one observation from dominating the strategy. The gap term raises risk when the parties remain far apart under the opponent model. The outer clipping keeps all downstream decisions on a common normalized risk scale.

6. Evaluation

During development, T2Agent was exercised against evaluation opponents designed to exhibit conceder, bouldware, cooperative-human, volatile, and hardliner behavior across multiple scenarios. Per-turn traces of risk, proposal target, estimated opponent utility, and realized own utility were used to verify that classification changes the concession curve in the intended direction.

7. Lessons and Suggestions

Experiments during development showed that asking a small language model to directly estimate risk and detailed preferences produced unstable outputs and timeouts. The final design therefore derives strategic signals from observable offer sequences and restricts the model to surface realization. This separation makes failures easier to diagnose and preserves deterministic behavior.

The frequency-and-stability opponent model remains an approximation and may not fully represent non-additive preferences or deliberately misleading behavior. Future work should investigate how to infer a more accurate opponent model from text without substantially increasing computational cost, while also improving the validation of the strategy’s thresholds.

Conclusions

T2Agent combines profile-adaptive breakdown risk, controlled utility targets, efficient non-repetitive bidding, and next-bid-aware acceptance. Its strategic core is independent of language-model output, while concise communication is protected by validation, a finite timeout, and deterministic fallback text.